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16 June 1978
SUPERSEDING
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10 July 1974

MILITARY SPECIFICATION

# CARRIER, CARGO, FULL-TRACKED, M548; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 Scope. This specification covers the processing of the Carrier, Cargo, Full-Tracked, M548; for storage outside of buildings, for immediate use shipment, and for domestic or overseas shipment, including carloading; meeting the requirements of Levels A and B processing (see 1.2).

* 1.2 Classification. Processing shall be of the following levels, as specified (see 6.2):

Level A - Maximum military protection.

Level A is the processing required for the protection of vehicle during shipment, handling, and storage exceeding 90 days from date of actual processing. This level does not provide for driveaway capability. It does provide for domestic or overseas shipment, including open deck loading.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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Level B - Minimum military protection.

Level B is the limited processing required for the protection of vehicle during shipment, handling and storage not to exceed 90 days from date of actual processing. This level provides for driveaway capability, when specified, and domestic or overseas shipment (excluding open deck loading).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

* SPECIFICATIONS

FEDERAL

| | |
|---|---|
| O-I-490 | - Inhibitor, Corrosion, Liquid Cooling System. |
| O-S-576 | - Sodium Bicarbonate, Technical. |
| P-D-220 | - Detergent, General Purpose. |
| GG-T-250 | - Tester, Antifreeze Solutions. |
| QQ-S-698 | - Steel, Sheet and Strip, Low-carbon. |
| QQ-S-781 | - Steel Strapping, Flat and Seals. |
| RR-W-360 | - Wire Fabric, Industrial. |
| TT-I-1795 | - Ink, Marking, Stencil, Opaque (Porous and Nonporous Surfaces). |
| TT-V-121 | - Varnish, Spar, Water-resisting. |
| UU-P-268 | - Paper, Kraft, Wrapping. |
| UU-T-81 | - Tag, Shipping and Stock. |
| VV-F-800 | - Fuel Oil, Diesel. |
| VV-L-800 | - Lubricating Oil, General Purpose, Preservative, (Water-displacing, Low Temperature). |
| ZZ-T-416 | - Tire, Pneumatic, Retread and Repair Materials. |
| MMM-A-178 | - Adhesive, Paper Label, Water-resistant. |
| PPP-B-601 | - Box, Wood, Cleated Plywood. |
| PPP-B-621 | - Box, Wood, Nailed and Lock-corner. |
| PPP-B-636 | - Box, Fiberboard. |
| PPP-T-42 | - Tape, Packaging/masking, Paper. |
| PPP-T-60 | - Tape, Pressure-sensitive Adhesive, Waterproof, for Packaging. |
| PPP-T-97 | - Tape, Pressure-sensitive Adhesive, Filament Reinforced. |

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| MIL-P-116 | - Preservation-packaging, Methods of. |
| MIL-B-117 | - Bag, Sleeve and Tubing - Interior Packaging. |



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| MIL-S-207 | - Sulfuric Acid, Electrolyte, Packaging, Packing, and Marking for Shipment and Storage of. |
| MIL-C-450 | - Coating Compound, Bituminous Solvent Type, Black (for Ammunition). |
| MIL-L-2105 | - Lubricating Oil, Multipurpose. |
| MIL-C-5501 | - Cap and Plug, Protective Dust and Moisture Seal. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-B-11188 | - Battery, Storage, Lead-acid. |
| MIL-A-11755 | - Antifreeze, Arctic Type. |
| MIL-B-12841 | - Basic Issue Items for Military Vehicles, Carriages and Equipment, Preparation for Shipment and Storage of. |
| MIL-P-13983 | - Paint, Temporary, Lusterless, Gasoline Removable. |
| MIL-C-16173 | - Corrosion Preventive Compound, Solvent Cut-back, Cold Application. |
| MIL-D-16791 | - Detergent, General Purpose (Liquid, Nonionic) |
| MIL-L-21260 | - Lubricating Oil, Internal Combustion Engine, Preservative and Break-in. |
| MIL-T-22085 | - Tape, Pressure-sensitive Adhesive, Preservation and Sealing. |
| MIL-B-22191 | - Barrier Material, Transparent, Flexible, Heat Sealable. |
| MIL-P-46002 | - Preservative Oil, Contact and Volatile Corrosion Inhibited. |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-T-50036 | - Talc, Technical, Tl. |
| MIL-H-52079 | - Hose, Preformed, Semiflexible, Reinforced. |
| MIL-D-81298 | - Dye, Liquid Red, for the Detection of Leaks in Aircraft Fuel Systems. |

STANDARDS

MILITARY

| | |
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| MIL-STD-129 | - Marking, for Shipment and Storage. |
| MS35842 | - Clamp, Hose, Low Pressure. |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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ASSOCIATION OF AMERCAN RAILROADS PUBLICATIONS

Section No. 1 - General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6 - Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, IL 60605.)

DEPARTMENT OF TRANSPORTATION

Hazardous Materials Regulations

(Application for copies should be addressed to the Department of Transportation, Hazardous Materials Regulations Board, Washington, DC 20590.)

DEPARTMENT OF TRANSPORTATION

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Department of Transportation Commission, Bureau of Motor Carrier Safety, Washington, DC 20590.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Level A.

* 3.1.1 One of the first ten production processed vehicles. Unless otherwise specified (see 6.2), one of the first ten production processed vehicles shall be subjected to the inspection and tests specified (see 4.2). Approval of one of the first ten production processed vehicles shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Except as otherwise specified by the procuring activity, any change to materials or design after approval of one of the first ten production processed vehicles shall require an additional production processed vehicle be inspected as specified (see 4.2). Government representative shall select the vehicle to be inspected.



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3.1.2 Government furnished equipment. Unless previously accomplished, government furnished equipment (GFE) (other than installed) shall be packaged, packed and marked in accordance with the individual document for the specific item. GFE shall be stowed with basic issue items (BII).

3.1.3 Preservatives and atomized spray equipment. When atomized spraying of preservative oils is specified, equipment shown in figures 2 and 3, or equivalent, shall be used. Compressed air supply lines shall be equipped with moisture separators every 50 feet or fraction thereof.

3.1.4 Processing records. Records of vehicle processing shall be maintained and shall be readily available for review by Government representative.

3.1.5 Disassembly. Projecting parts whose removal will accomplish desired cube reduction and parts susceptible to damage and pilferage shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Removed parts shall be preserved, packaged, and packed. Packed parts shall be identified and stowed securely within the vehicle.

* 3.1.5.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to Type A of UU-T-81. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178.

* 3.1.6 Record forms. Two copies of DD Form 1397 shall be provided. Information on forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 (see 6.4) shall be placed in a bag conforming to Type I, Class B, Style 2 of MIL-B-117, 6 mil; bag shall be closed by heat sealing and securely attached in driver's compartment of vehicle. The other copy of DD Form 1397 shall be waterproofed with adhesive conforming to MMM-A-178 or sealed in a bag and securely attached in a conspicuous location on the exterior of the vehicle.

3.1.7 Cleaning and drying.

3.1.7.1 Interior of vehicle. Interior surfaces of vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or Type 1 of MIL-D-16791 in warm water. Water or other liquid under pressure or steam cleaning shall not be used. After cleaning, cleaned surfaces shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing to assure that no liquids enter instruments, connections, or other components susceptible to water damage and that water does not accumulate in areas where it cannot drain or be dried.



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3.1.7.1.1 Cleaning and drying of battery supports and retainers. Battery supports and retainers shall be cleaned with a solution of one-half pound of sodium bicarbonate conforming to O-S-576 per gallon of water. After cleaning, cleaned surfaces shall be flushed with clean water, then thoroughly dried. Dried surfaces shall then be preserved (see 3.1.8.2).

3.1.7.1.2 Cleaning and drying of backrests and seats. Backrest and seat cushions shall be cleaned with a solution of detergent conforming to P-D-220, or Type 1 of MIL-D-16791, in warm water. After cleaning, cushions shall be wiped with cloths saturated with clean water to remove cleaning solution. Care shall be taken not to saturate the cushions with cleaning solution or water. After rinsing, cushions shall be dried, then protected (see 3.1.9.3).

3.1.7.2 Exterior of vehicle. Exterior of vehicle, including cargo bed, shall be cleaned using a solution of detergent conforming to P-D-220, or Type 1 of MIL-D-16791, in warm water or steam. Cleaning shall remove all foreign matter. After cleaning, surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into driver's or engine compartments.

3.1.8 Preservation.

3.1.8.1 Relubrication. Vehicle shall be lubricated using materials conforming to drawings, specifications, or lubrication orders applicable to the vehicle, after the vehicle has been driven more than 50 miles or has been exposed to contaminating environments since initial lubrication which necessitates cleaning of vehicle exterior as specified (see 3.1.7.2).

3.1.8.2 Preservation of battery supports and retainers. Top battery supports and retainers shall be preserved with compound conforming to MIL-C-450.

3.1.8.3 Transmission, transfer assembly, control differential, and final drives. Transmission shall contain lubricating oil conforming to Type 1, Grade 1 of MIL-L-21260 filled to operating level. Transfer assembly, control differential, and final drives shall contain lubricating oil conforming to Type 1, Grade 10 or 30, as applicable, of MIL-L-21260 filled to operating level. DD Form 1397 shall be annotated with type and grade of lubricant used.

* 3.1.8.4 Cooling system. Cooling system shall be protected by one of the following procedures:

a. For shipment to, and storage in, areas where the temperature drops below minus 40 degrees Fahrenheit (-40°F), systems shall be protected as



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specified in 3.1.8.4.3.

b. For shipment and storage within the bounds of 30 degrees north latitude and 20 degrees south latitude, except continental United States, systems shall be protected as specified in 3.1.8.4.2.

c. For all other shipments, cooling systems shall be protected as specified in 3.1.8.4.1.

NOTE: DD Form 1397 (see 3.1.6) shall be completed to indicate coolant used.

* 3.1.8.4.1 Water and antifreeze procedure. Cooling system shall be filled to operating level with a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to MIL-A-46153, and water. Engine shall be operated until a temperature has been reached that causes the thermostat to open, to assure complete mixing and even distribution of the antifreeze solution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN," shall be securely attached to radiator filler neck.

* 3.1.8.4.2 Water and corrosion inhibitor procedure. Cooling system shall be filled with clear water up to, but not including, the radiator upper tank. A corrosion inhibitor conforming to O-I-490 shall be added in the proportion of five ounces of the inhibitor for each ten quarts of water. The inhibitor shall be dissolved in two quarts of warm water and poured into the radiator while the engine is idling. More water shall be added, if necessary, to fill the radiator to operating level. A warning tag, bearing the information "COOLING SYSTEM DOES NOT CONTAIN ANTIFREEZE - FILLED WITH WATER AND INHIBITOR," shall be securely attached to radiator filler neck.

* 3.1.8.4.3 Antifreeze compound procedure. Cooling system shall be filled to operating level with antifreeze compound conforming to MIL-A-11755. The compound shall be used without dilution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH ANTIFREEZE (ARCTIC-TYPE) - DO NOT DRAIN," shall be securely attached to radiator filler neck.

* 3.1.8.5 Crankcase. Crankcases of compression ignition engines shall be filled to operating level with lubricating oil conforming to Type 1 of MIL-L-21260 of the seasonal grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the type and grade of lubricant used.

3.1.8.6 Compression ignition engine. Compression ignition engine preservation shall be as specified (see 3.1.8.6.1 and 3.1.8.6.2) in an uninterrupted sequence.



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* 3.1.8.6.1 Fuel system and combustion chamber. Prior to processing engine, the engine shall be cooled to assure that cylinder head temperature, measured at injector nozzle flange surface of all cylinders, is not more than 100°F. Cooling shall be accomplished by induced air currents, circulation of engine coolant, or by waiting the period of time required to arrive at the above specified temperature. When ambient temperature exceeds 100°F, engine shall be cooled to a temperature equivalent to the ambient. After the engine has been cooled, the fuel supply system from the fuel tank shall be shut off. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with lubricating oil conforming to Grade 1 of MIL-P-46002 to which has been added an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with diesel fuel conforming to VV-F-800. (The container shown in figure 1 has proven satisfactory for engine preservation.) Disconnect the fuel line between the primary fuel filter and fuel pump at the primary fuel filter outlet. Drain the residual diesel fuel from the secondary fuel filter. Remove filter can and remove filter element. Fill filter can with oil, lubricating, Grade 1, MIL-P-46002, and replace. Connect fuel line from auxiliary fuel container to fuel pump supply line. Disconnect residual vehicle fuel return line at quick disconnect coupling. Connect a transparent plastic fuel line to the engine end of the disconnected fuel return line, insert the other end of plastic fuel line into a recovery container to collect the residual returned oil. The fuel valve on the auxiliary fuel tank shall be turned to the lubricating oil position, Grade 1, MIL-P-46002. Engine shall be started and operated not more than 1,200 revolutions per minute until the MIL-P-46002 oil with red dye is observed flowing through the plastic fuel return line. Engine shall be stopped at the instant the dyed oil is observed. Disconnect air cleaner hose from the air inlet elbow. An air restrictor boot shall be installed on air inlet elbow and secured to assure that air to the engine is completely cut off. (The air restrictor boot shown in figure 5 has proven satisfactory for engine preservation.) Place engine fuel control in RUN position. Crank engine with starter (NOTE: Engine may fire for approximately five seconds) for a minimum period of 30 seconds and not exceeding a maximum of 45 seconds. The fuel valve on the auxiliary fuel container shall be turned to the off position. Remove the filter can from the secondary fuel filter. Drain the oil, lubricating, Grade 1, MIL-P-46002, from can. Wipe clean with lint free cloth. Replace element, fill can with diesel fuel, VV-F-800, and replace filter can.

CAUTION: Special precautions shall be taken to assure that time limits specified shall not be exceeded, or the engine starter or starter solenoid may be damaged. Also, excessive cranking could cause hydrostatic lock.



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3.1.8.6.2 Engine purging. Purge fuel return lines of all preservative oil, Grade 1, MIL-P-46002, by turning the auxiliary fuel tank valve to the diesel fuel position and closing engine fuel control shut off. Crank engine with starter for a period of time to remove all traces of red colored lubricating oil from the plastic return line.

CAUTION: Do not operate starter for a period exceeding 45 seconds.

If preservative oil has not been purged, rest starter for three minutes and repeat purge procedure. Turn the auxiliary fuel container valve to the off position. Remove fuel line and reconnect to the outlet side of primary fuel filter. Turn on vehicle fuel supply system. Remove air restrictor boot. Disconnect plastic fuel return line, and reconnect vehicle fuel return at the quick disconnect coupling (see 4.5.2.4).

* 3.1.8.6.3 Preservation through air intake and exhaust system. After preservation in accordance with 3.1.8.6.1 and 3.1.8.6.2, one ounce of preservative oil conforming to Grade 1 of MIL-P-46002 shall be atomized into exhaust opening. Opening shall then be sealed with tape conforming to Type IV, Class 1 of PPP-T-60 or Type II of MIL-T-22085. The hose shall be disconnected at the air intake, and one ounce of preservative oil, conforming to Grade 1, MIL-P-46002, shall be atomized into the air intake opening. The air intake opening and the engine crankcase breathers shall then be sealed with plastic plugs conforming to MIL-C-5501 or with tape conforming to Type IV, Class 1 of PPP-T-60 or Type II of MIL-T-22085. A red warning tag bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL HOSE" shall be placed in a conspicuous location within driver's compartment. DD Form 1397 shall be annotated to show engine is preserved with VCI and oil.

* 3.1.8.6.4 Preservation through oil level gage rod opening. After preservation in accordance with 3.1.8.6.1 and 3.1.8.6.2, the oil level gage rod shall be removed, and six ounces of preservative oil conforming to Grade 1 of MIL-P-46002 shall be atomized into the crankcase through the oil level gage rod opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. Spray nozzle shall not be submerged in the crankcase oil. After spraying has been accomplished, oil level gage rod shall be reinstalled. All openings to the engine interior, including oil level gage rod opening and oil filler cap, shall be sealed with tape conforming to Type IV, Class 1 of PPP-T-60 or Type II of MIL-T-22085.

3.1.8.6.5 Preservation through flywheel housing. Two ounces of preservative oil conforming to Grade 1 of MIL-P-46002 shall be atomized into the flywheel housing.



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* 3.1.8.6.6 Air cleaner. Air cleaner openings shall be sealed with tape conforming to Type IV, Class 1 of PPP-T-60 or Type II of MIL-T-22085.

* 3.1.8.7 Personnel and engine compartment heaters and lines. Personnel and engine heaters shall have the fuel supply shut off valve located at the inlet side of fuel filters turned to the OFF position. The main fuel line supplying fuel to the heaters shall be disconnected at a point closest to shut off valves. Fuel from the fuel lines shall be allowed to drain. Seal ends of disconnected fuel lines and shut off valves with plastic plugs/caps conforming to MIL-C-5501 or with tape conforming to Type IV, Class 1 of PPP-T-60, or Type II of MIL-T-22085. External exhaust opening shall have opening sealed with tape conforming to Type IV, Class 1 of PPP-T-60, or Type II of MIL-T-22085. Prior to placing personnel or engine heaters in operation, remove plugs/caps or tape from fuel lines, exhaust stack and shut off valves. Operate heater fuel pump and drain a minimum of one quart of fuel through the fuel lines into a portable container. Reconnect heater fuel lines. Four warning tags, each bearing the information "HEATER FUEL LINES DISCONNECTED AND SEALED - REMOVE SEALS, RECONNECT FUEL LINES PRIOR TO STARTING HEATERS," shall be prepared. One tag each shall be secured to the personnel and engine heater operating switch, one each to the personnel and engine heaters.

3.1.8.8 Fuel tank preservation. Fuel tank shall be drained to the maximum extent possible. Fuel tank cap and filler screen shall be removed and coated with lubricating oil conforming to Type 1, Grade 30 of MIL-L-21260. One quart of lubricating oil conforming to Type 1, Grade 10 of MIL-L-21260 shall be added to each five gallons or portion thereof of fuel that remains after draining tank. Tank cap and filler screen shall be reinstalled (see 4.5.2.2).

3.1.8.8.1 Fuel tank security. After processing the fuel tank as specified (see 3.1.8.8), the fuel cap shall be secured with a bracket fabricated and installed in accordance with figure 6.

* 3.1.8.9 Winch assembly. Winch gear case shall be filled to operating level with lubricant conforming to Grade 90 of MIL-L-2105. Wire cable shall be unreeled and all its surfaces coated with preservative conforming to Grade 1 of MIL-C-16173. The cable shall be rewound. While the cable is being rewound, it shall be inspected for damage to the preservative coating. Any breaks in coating shall be represerved prior to rewinding. All other moving, mating parts shall be coated with grease conforming to MIL-G-10924.

* 3.1.8.10 Doors and tailgate. Rubber seals around doors and tailgate shall be coated with powdered talc conforming to Type IV, Class C of ZZ-T-416 or talc, technical, MIL-T-50036.

* 3.1.8.11 Ventilation.



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* 3.1.8.11.1 Engine compartment access plate. Engine compartment access plate and gasket, attaching hardware, and drain plugs shall be removed for ventilation. Bare metal surfaces of drain plugs shall be preserved with compound conforming to Grade 4 of MIL-C-16173. Plate, gasket, preserved drain plugs, and four access plate mounting screws and washers shall be packaged as specified in 3.1.9.6. A screen conforming to figure 4 shall be installed in access plate opening and secured with four of the existing mounting screws and washers. The information "REMOVE SCREEN, INSTALL ACCESS PLATE AND GASKET AND DRAIN PLUGS BEFORE VEHICLE OPERATION" shall be stenciled on the exterior of the vehicle using white or yellow paint conforming to MIL-P-13983. Stenciling shall be in characters not less than 3/4 inch high.

* 3.1.8.12 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted metal surfaces on the exterior of the vehicle, except the track shoes, shall be coated with compound conforming to Grade 1 of MIL-C-16173. All exposed, unpainted, unplated metal surfaces on the interior of the vehicle shall be coated with compound conforming to Grade 4 of MIL-C-16173.

3.1.9 Packaging.

3.1.9.1 Dry charged batteries and cables. Dry charged batteries shall be installed and secured in the vehicle battery carrier. Battery cables shall be secured to battery carrier with 3/4-inch tape conforming to Type IV of PPP-T-97. Battery filler cap openings shall be sealed by placing a 2-inch wide by 3-mil thick piece of film conforming to Type II of MIL-B-22191 over each filler cap opening with cap removed. The sheet shall be of sufficient length to allow it to be depressed into the opening to the same depth as the filler plug. Filler caps shall be screwed or inserted into openings to form a complete seal without damaging the sheet. If batteries have been processed in accordance with MIL-B-11188, they need not be reprocessed as above.

3.1.9.2 Electrolyte. Electrolyte shall be packaged, packed, and marked as specified for Type IV, Class 1, in accordance with MIL-S-207. The packed electrolyte shall be stowed with BII and secured independently to permit separate removal.

3.1.9.3 Packaging of backrests and seats. Cushions of backrests and seats (see 3.1.7.1.2) shall be covered with paper conforming to Grade B of UU-P-268 with a minimum basis weight of 60 pounds. Paper shall be secured with tape conforming to Type I of PPP-T-42.

3.1.9.4 Fire extinguishers. Fire extinguishers shall contain a minimum of 90 percent of rated full charge. All seals shall be intact. DA Form 253 shall be completed and attached securely to each extinguisher (see 6.3).



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3.1.9.5 Basic issue items. BII, as applicable, shall be packaged and packed in accordance with MIL-B-12841 or other documents designated by the responsible agency. BII shall be stowed and secured as specified in 3.1.9.5.1.

* 3.1.9.5.1 Stowage and securement of BII. BII and items removed for shipment shall be identified to pertinent vehicle by serial number and except during shipment, shall be stored inside buildings. Packed BII and removed items shall be placed within the cargo compartment of vehicle. Large wooden boxes shall be placed on the vehicle floor and shall be secured with 1-1/4 inch wide strapping conforming to Type I, Class B of QQ-S-781. Strapping shall be secured to holding devices within the compartment. Smaller wooden boxes shall be secured to the larger boxes with 3/4 inch wide strapping conforming to Type I, Class B of QQ-S-781. All boxes shall be secured in such a manner as to prevent any movement during transit and to prevent damage to containers or vehicle interior. Corner protectors shall be used under all strapping.

* 3.1.9.6 Access plate, gasket, and drain plugs. Access plate, gasket, and preserved drain plugs (see 3.1.8.11) shall be packaged in a box conforming to Type CF, Class Weather-resistant of PPP-B-636. Box shall be closed with tape conforming to Type III, Class 1 of PPP-T-60, identified as to contents, and securely stowed within crew compartment.

* 3.1.9.7 Tow hooks and tow pintle. Tow hooks, pintle, and related hardware shall be removed for shipment and packaged in a Type CF, Class Weather-resistant box conforming to PPP-B-636. Box shall be closed with tape conforming to Type III, Class 1 of PPP-T-60, identified as to contents, and securely stowed within crew compartment.

3.1.9.8 Exhaust pipe. The exhaust pipe and attaching clamps shall be removed for shipment and packaged in a box conforming to Type CF, Class Weather-resistant of PPP-B-636. Box shall be closed with tape conforming to PPP-T-60, Type III, Class 1, identified as to contents, and securely stowed within crew compartment.

* 3.1.9.9 Cab cover, windshield, and door windows. Cab cover, windshield and door windows shall be removed for shipment and storage. Unpainted metal surfaces shall be coated with preservative conforming to MIL-C-16173, Grade 4. Removed bolts, nuts, screws, pins and washers shall be placed in one of the mating parts and secured. Fabric cover shall be thoroughly dried, folded and placed in container conforming to Type CF, Class Weather-resistant of PPP-B-636. Box shall be closed with tape conforming to Type III, Class 1 of PPP-T-60. Windshield wipers shall be placed in a bag conforming to MIL-B-117, Type I, Class B, Style 2, 4 mil. Rubber molding around windows shall be dusted with talc conforming to Type IV, Class C of ZZ-T-416 or talc, technical, MIL-T-50036. Packaged cab cover, windshield and door windows shall be packed in a box conforming to Class 2 of PPP-B-621 or optional PPP-B-601, Style I. Items shall



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be blocked, braced and cushioned in container. Box shall be identified and stowed with BII.

* 3.1.9.10 Cargo cover installation. Cargo cover installation, except tie down hooks, shall be removed for shipment. Unpainted metal surfaces shall be coated with preservative conforming to MIL-C-16173, Grade 4. Removed screws, nuts and washers shall be packaged separately in bag conforming to MIL-B-117, Type I, Class B, Style 2, 4 mil, and placed within cover assembly container. Cover assembly with ropes shall be thoroughly dried, folded and placed into container conforming to Type CF, Class Weather-resistant of PPP-B-636. Box shall be closed with tape conforming to Type III, Class 1 of PPP-T-60, then placed with bows in wood box conforming to Class 2 of PPP-B-621 or optional PPP-B-601, Style I. Items shall be blocked, braced and cushioned in container. Box shall be identified and stowed with BII.

3.1.10 Vehicle closure.

3.1.10.1 Closure kit. Unless otherwise specified (see 6.2), each vehicle shall be provided with a protective closure kit fabricated and installed in accordance with Appendix A and figure 7.

* 3.2 Level B. Vehicles shall be processed in the same manner as specified for Level A, with the following exceptions:

3.2.1 Transmission, transfer assembly, control differential, and final drives. Transmission, transfer assembly, control differential, and final drives shall contain operational lubricant as specified on applicable drawings, specifications, or lubrication orders, filled to operating level. If, however, these units contain lubricating oil conforming to Type I, Grade 10 or 30 of MIL-L-21260, an additional amount of the same oil shall be added to attain operating level. Operating lubricants shall not be mixed with MIL-L-21260. DD Form 1397 shall be annotated to indicate grade of lubricant or preservative oil used.

3.2.2 Engine crankcase. Engine crankcase shall contain normal operational lubricant as specified in lubrication order, filled to operational level. DD Form 1397 shall be annotated to indicate grade of lubricant used (see 3.1.8.5).

3.2.3 Engine preservation. The engine shall require no preservation for Level B shipment and storage (see 3.1.8.6).

3.2.4 Fuel tank. Unless otherwise specified (see 6.2), vehicles shall be shipped without draining residual fuel from the fuel tanks (see 3.1.8.8).

3.2.5 Personnel heater and fuel pump. Unless otherwise specified (see 6.1), personnel heaters and fuel pumps shall be in ready-to-use condition (see 3.1.8.7).



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3.2.6 Winch assembly. Winch gear shall be filled with operational lubricant. Cable shall not be preserved (see 3.1.8.9).

* 3.2.7 Backrests and seats. Cushions of backrest and seats may be covered or uncovered (see 3.1.9.3).

* 3.2.8 Tow hooks and tow pintle. Tow hooks and tow pintle shall be removed for overseas shipment only (see 3.1.9.7).

* 3.2.9 Exhaust pipe. Exhaust pipe and clamps shall be removed and packaged for overseas shipment only (see 3.1.9.8).

* 3.2.10 Vehicle closure. Vehicle closure kit shall not be provided for Level B shipment and storage. Opening to vehicle interior shall be sealed in accordance with Appendix B and figure 8 (see 3.1.10.1).

* 3.3 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Material shall be free from all defects and inperfections that might affect the serviceability and appearance of the finished product (see 4.5.1 and 6.5).

3.4 Loading.

3.4.1 Loading flat cars. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of Section 1, General Rules Governing the Loading of Commodities on Open Top Cars, and figure 87 or 87A, Section 6, Rules Governing the Loading of Department of Defense Material on Open Top Cars, of the Association of American Railroad publication.

3.4.2 Highway shipment. Loading of vehicles for shipment by haulaway and rules for shipment by driveaway or towaway shall be in accordance with Department of Transportation publication, "Federal Motor Carrier Safety Regulations," and applicable military publications.

3.4.3 Reprocessing engine after loading - Level A. If engine is operated in connection with the moving of vehicle to loading area or during the loading itself, the engine shall be reprocessed as specified (see 3.1.8.6 through 3.1.8.6.6 and 3.1.8.8).

3.4.3.1 Reprocessing engine after loading - Level B. If engine is operated in connection with movement of vehicle for loading or unloading, there shall be no additional processing of engine.

3.5 Marking. In addition to any special marking required in the contract (see 6.2), vehicle shall be marked in accordance with MIL-STD-129.

3.5.1 Lifting points. The legend "LIFT HERE" with arrow pointing to the lifting eye shall be stenciled adjacent to each lifting eye using white ink conforming to TT-I-1795.



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3.6 Drive-on - drive-off capability. When vehicles are to be operated for loading or unloading, the following provisions shall apply:

3.6.1 Fuel tanks. As specified (see 6.2), additional fuel shall be added, as required, to accomplish movement of vehicle.

3.6.2 Batteries and electrolyte. New batteries with electrolyte added and fully charged shall be installed in the vehicle and battery cables properly connected.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Inspection records. Contractor shall maintain records of all inspections performed and such records shall be readily available for review by the Government representative.

4.2 One of the first ten production processed vehicles. One of the first ten production processed vehicles (see 3.1.1) shall be subjected to the inspection and tests specified (see 4.5).

4.3 Production processed vehicles. Unless otherwise specified (see 6.2), all production processed vehicles shall be subjected to the inspections and tests specified (see 4.5.2 through 4.5.2.3).

* 4.4 Rejection. Failure of one of the first ten, or any production processed vehicle, to conform to the applicable requirements of this specification shall be cause for rejection of the vehicles by the Government. No vehicles shall be accepted until objective evidence that the contractor has corrected the condition causing the rejection has been provided to the Government.

4.5 Quality conformance inspection.

* 4.5.1 Materials. All materials to be used in processing of vehicles shall be inspected in accordance with the material specification or certified inspection and laboratory test reports shall be provided which show that furnished materials conform to the detailed specifications (see 3.3 and 6.5). When materials are listed on a Qualified Products List (QPL), they shall be obtained from one of the approved sources indicated.



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* 4.5.2 Processing. Except as otherwise specified herein, vehicle processing shall be inspected to determine conformance to this specification. Inspection of processing shall include all items specified in table I, and as specified below.

* 4.5.2.1 Cleaning. To determine conformance to 3.1.7.1, interior of vehicles shall be examined for cleanliness. All vehicles shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.1.7.2, exterior of vehicle shall be examined for cleanliness. Surfaces to which tape is to be applied shall be examined for cleanliness before application.

4.5.2.2 Fuel tank. To determine conformance to 3.1.8.8, visual inspection of fuel tank interior shall assure that complete specified processing has been accomplished.

* 4.5.2.3 Cooling system. To determine conformance to 3.1.8.4, the engine coolant shall be tested using a hydrometer-thermometer type tester, with a range of minus 60°F to plus 160°F, conforming to GG-T-250.

4.5.2.4 Engine. To determine conformance to 3.1.8.6.1 through 3.1.8.6.6, as applicable, interior of the engine from one of the first ten production processed vehicles shall be examined for surface coverage. Engine shall be disassembled to the extent necessary to permit visual examination of all surfaces within the combustion chamber. The combustion chamber is all surfaces within the cylinder from and including the surface of the head within the cylinder. All surfaces within the combustion chamber shall have a "wet" coating of preservative oil such as is obtained when an item is dipped or flushed with the oil. The processing method of the approved preserved engine shall be submitted in writing for use in processing engines in all subsequent production vehicles.

* 5. PACKAGING

This section is not applicable to this specification.

6. NOTES

* 6.1 Intended use. This specification covers processing of the M548 vehicle for storage outside of buildings, for immediate shipment and domestic or overseas shipment, including carloading, (see 1.1).

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level of processing (see 1.2).



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c. If inspection of one of the first ten production processed vehicles is not required (see 3.1.1).
d. Applicable procedure for protection of cooling system (see 3.1.8.4).
e. If vehicle closure kit is not required (see 3.1.10.1).
f. If inspection and tests other than those specified are required (see 4.3).
g. If additional fuel shall be supplied (see 3.6.1).
h. If special marking is required (see 3.5).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide ($CO_2$) fire extinguisher cylinders. Cylinders should not be dropped, be permitted to strike each other, or be handled roughly. Extreme care should be exercised during the reinstallation operation to avoid tripping fire extinguisher control system (see 3.1.9.4).

* 6.4 Forms. A copy of the "Equipment Log Book" and all required forms (see 3.1.6) will be furnished to the contractor by the Government at least 30 days before shipment of the vehicles required by the contract delivery schedule.

* 6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specification without jeopardizing the intended use of the item shall be encouraged (see 4.5.1).

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - AT

Preparing activity:
Army - AT

(Project 2320-AA48)



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*

TABLE I. Processing inspection.
(See indicated paragraphs for level A, B requirements)

| Component | Cleaning Levels A & B | Preservation | | Packaging/Stowage | |
|---|---|---|---|---|---|
| | | Level A | Level B | Level A | Level B |
| Processing records | | | | 3.1.4 | 3.1.4 |
| Disassembly | | | | 3.1.5 | 3.1.5 |
| Matchmarking | | | | 3.1.5.1 | 3.1.5.1 |
| Interior of vehicle | 3.1.7.1 | | | | |
| Battery supports & retainers | 3.1.7.1.1 | 3.1.8.2 | 3.1.8.2 | | |
| Backrests & seats | 3.1.7.1.2 | | | 3.1.9.3 | 3.2.7 |
| Exterior of vehicle | 3.1.7.2 | | | | |
| Relubrication | | 3.1.8.1 | 3.1.8.1 | | |
| Transmission, transfer assembly, control differential, & final drives 1/ | | 3.1.8.3 | 3.2.1 | | |
| Engine crankcase 1/ | | 3.1.8.5 | 3.2.2 | | |
| Compression ignition engine | | 3.1.8.6 | 3.2.3 | | |
| Preservation thru fuel system & combustion chamber | | 3.1.8.6 | 3.2.3 | | |
| Engine purging | | 3.1.8.6.2 | 3.2.3 | | |
| Preservation thru air intake & exhaust system 1/ | | 3.1.8.6.3 | 3.2.3 | 3.1.8.6.3 | |
| Preservation thru oil level gage rod opening | | 3.1.8.6.4 | 3.2.3 | 3.1.8.6.4 | |
| Preservation thru flywheel housing | | 3.1.8.6.5 | | | |
| Air cleaner | | | | 3.1.8.6.6 | |
| Personnel & engine compartment heaters & lines | | 3.1.8.7 | 3.2.5 | 3.1.8.7 | |
| Fuel tank | | 3.1.8.8 | 3.2.4 | | |
| Engine Cooling system 1/ | | 3.1.8.4 | 3.1.8.4 | | |
| Water & antifreeze procedure | | 3.1.8.4.1 | 3.1.8.4.1 | | |



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TABLE I. Processing inspection. - Continued
(See indicated paragraphs for level A, B requirements)

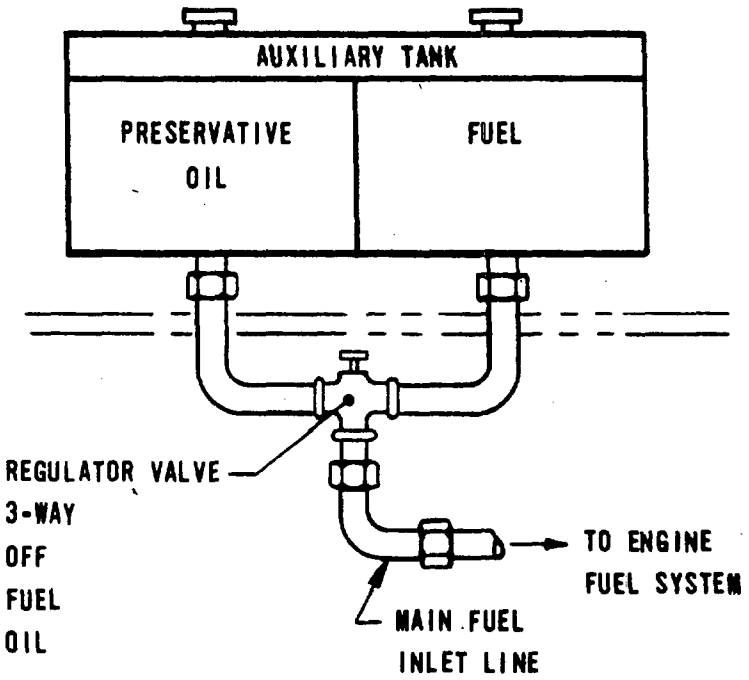
| Component | Cleaning Levels A & B | Preservation | | Packaging/Stowage | |
|---|---|---|---|---|---|
| | | Level A | Level B | Level A | Level B |
| Water & corrosion inhibitor procedure | | 3.1.8.4.2 | 3.1.8.4.2 | | |
| Arctic antifreeze procedure | | 3.1.8.4.3 | 3.1.8.4.3 | | |
| Winch assembly | | 3.1.8.9 | 3.2.6 | | |
| Doors & tailgate | | 3.1.8.10 | 3.1.8.10 | | |
| Ventilation & screen | | 3.1.8.11.1 | 3.1.8.11.1 | 3.1.9.6 | 3.1.9.6 |
| Miscellaneous preservation | | 3.1.8.12 | 3.1.8.12 | | |
| Dry charged batteries & cables | | 3.1.9.1 | 3.1.9.1 | 3.1.9.1 | 3.1.9.1 |
| Electrolyte | | | | 3.1.9.2 | 3.1.9.2 |
| Fire extinguishers | | | | 3.1.9.4 | 3.1.9.4 |
| Record forms | | | | 3.1.6 | 3.1.6 |
| Access plate, gasket & plugs | | | | 3.1.9.6 | 3.1.9.6 |
| Tow hooks & pintle | | | | 3.1.9.7 | 3.2.8 |
| Basic issue items | | | | 3.1.9.5 | 3.1.9.5 |
| Stowage & securement of BII | | | | 3.1.9.5.1 | 3.1.9.5.1 |
| Exhaust pipe | | | | 3.1.9.8 | 3.2.9 |
| Cab cover, windshield & door windows | | 3.1.9.9 | 3.1.9.9 | 3.1.9.9 | 3.1.9.9 |
| Cargo cover installation | | 3.1.9.10 | 3.1.9.10 | 3.1.9.10 | 3.1.9.10 |
| Vehicle closure | | | | 3.1.10 | 3.2.10 |
| Loading on flat cars | | | | 3.4.1 | 3.4.1 |
| Highway shipment | | | | 3.4.2 | 3.4.2 |
| Reprocessing engine after loading | | | | 3.4.3 | 3.4.3.1 |
| Marking | | | | 3.5 | 3.5 |
| Lifting points | | | | 3.5.1 | 3.5.1 |

1/ Inspect DD Form 1397





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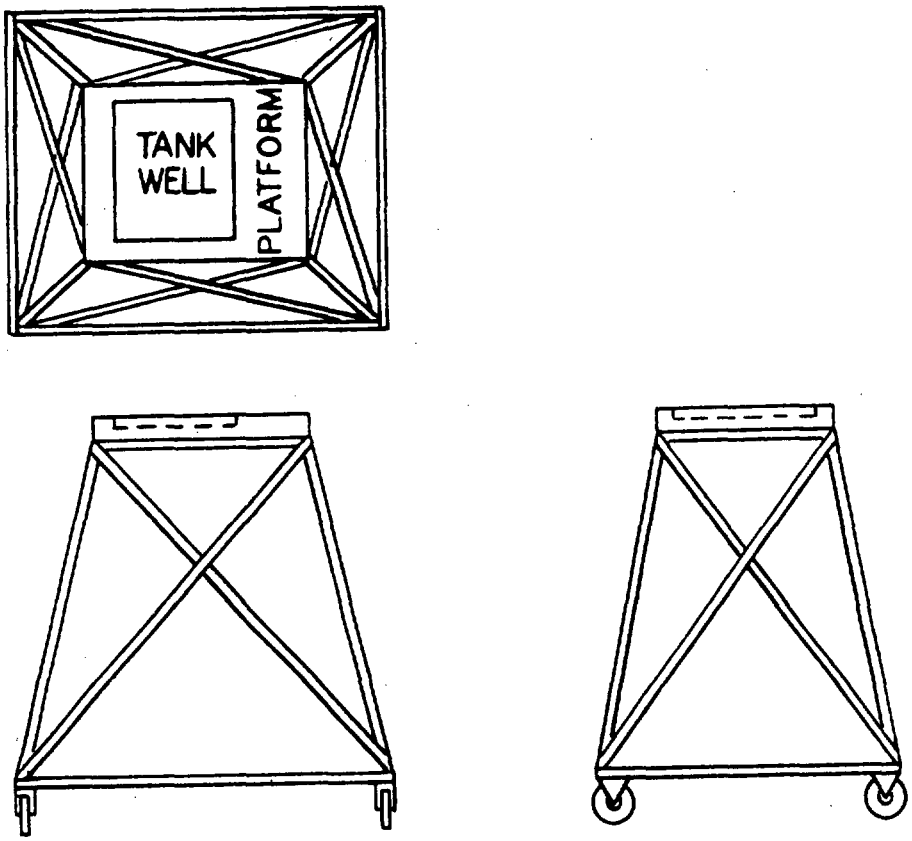



Sheet 1 of 2

FIGURE 1. Auxiliary fuel container.



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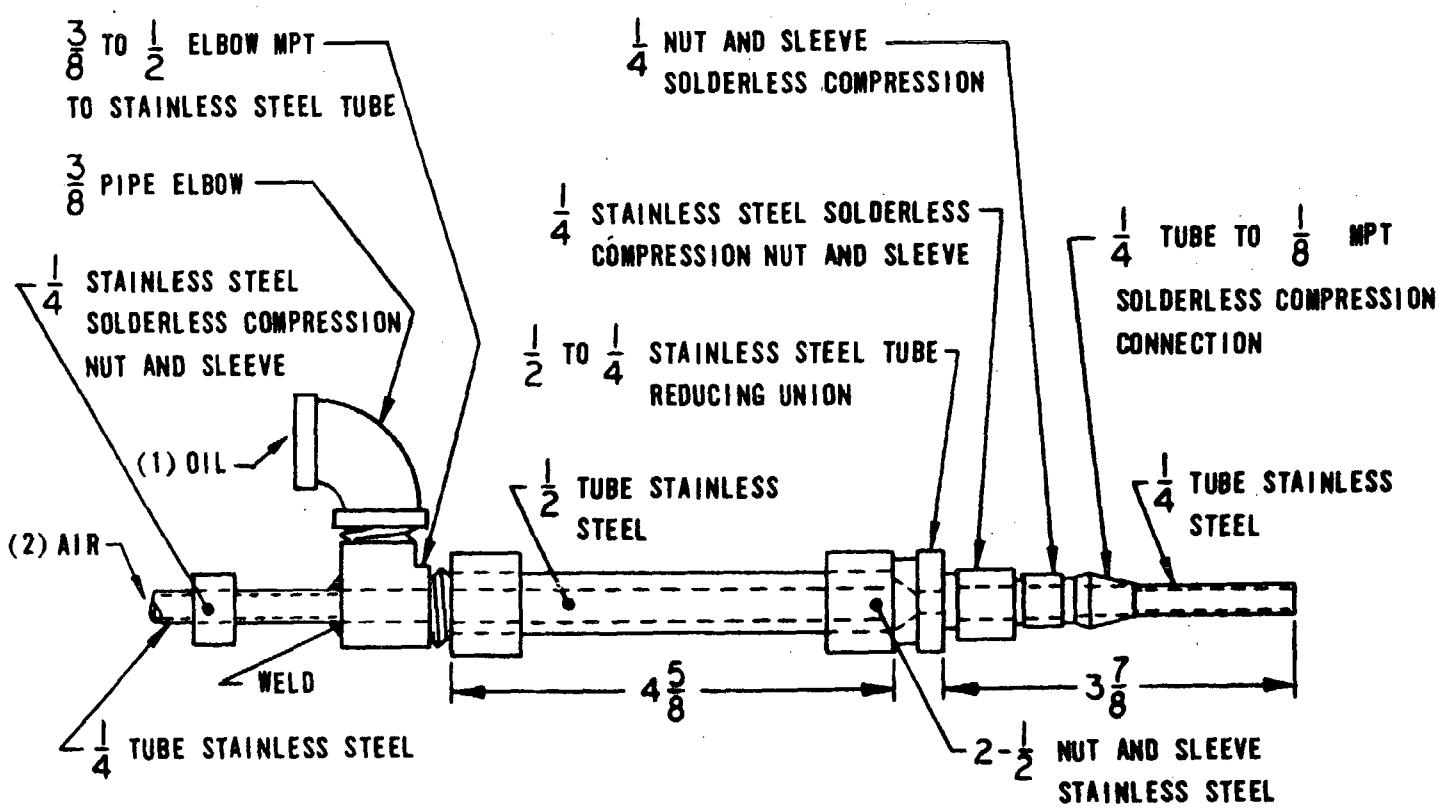



CONTAINER SUPPORT Sheet 2 of 2

FIGURE 1. Auxiliary fuel container.- Continued



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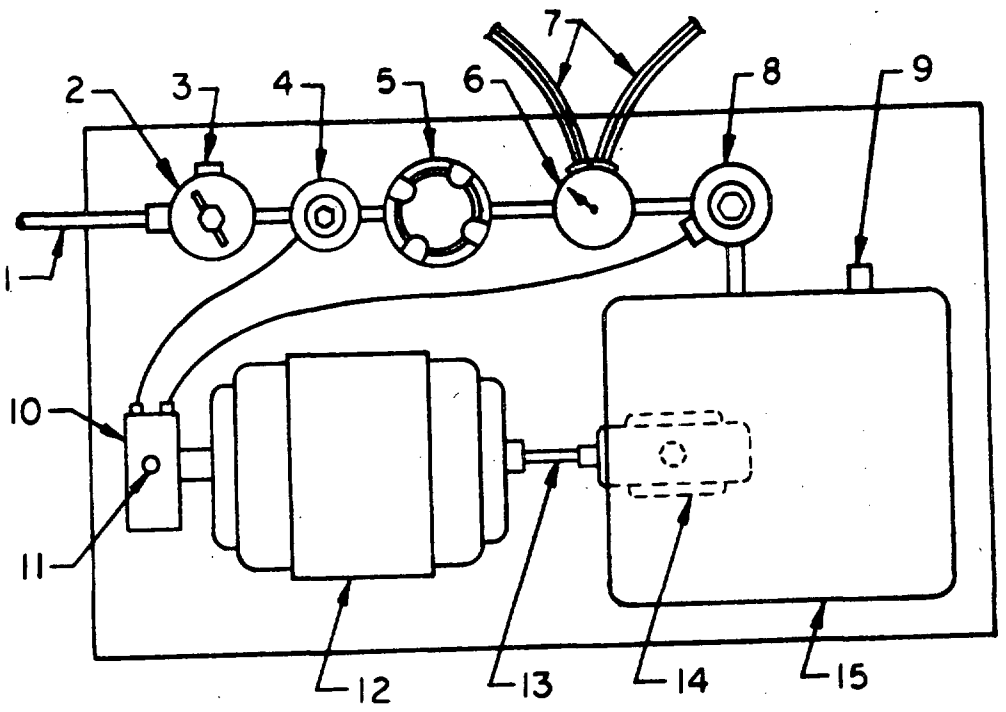

NOTE: (1) AND (2) CONNECT TO CORRESPONDING LINES ON FIGURE 3

FABRICATION DETAILS FOR OIL SPRAY ATOMIZING NOZZLE

FIGURE 2. Atomizing device.



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1. AIR LINE
2. AIR PRESSURE REGULATOR
3. AIR PRESSURE GAGE
4. SOLENOID VALVE
5. MOISTURE SEPARATOR
6. OIL PRESSURE GAGE
7. TWO DOUBLE TAPED LINES
8. SOLENOID VALVE
9. OIL TANK LEVEL GAGE
10. ELECTRIC JUNCTION BOX
11. ELECTRIC LINE
12. MOTOR 1/4 H P
13. SHAFT
14. POSITIVE DISPLACEMENT OIL PUMP
15. OIL TANK

NOTE:
THIS EQUIPMENT HAS PROVEN SATISFACTORY FOR ATOMIZED SPRAYING OF PRESERVATIVE OIL IN CONJUNCTION WITH FIGURE 2.

FIGURE 3. Pressure pump.



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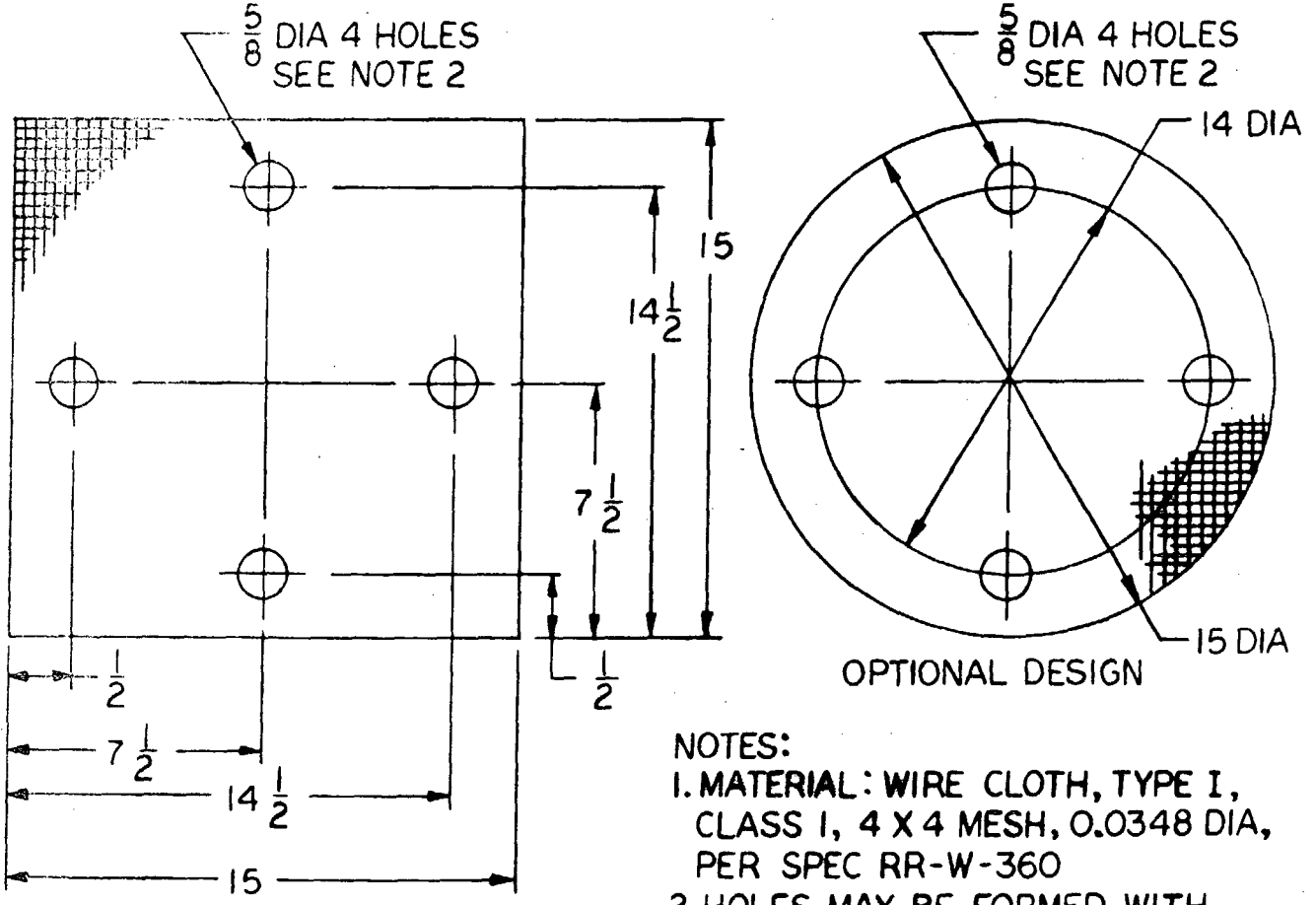
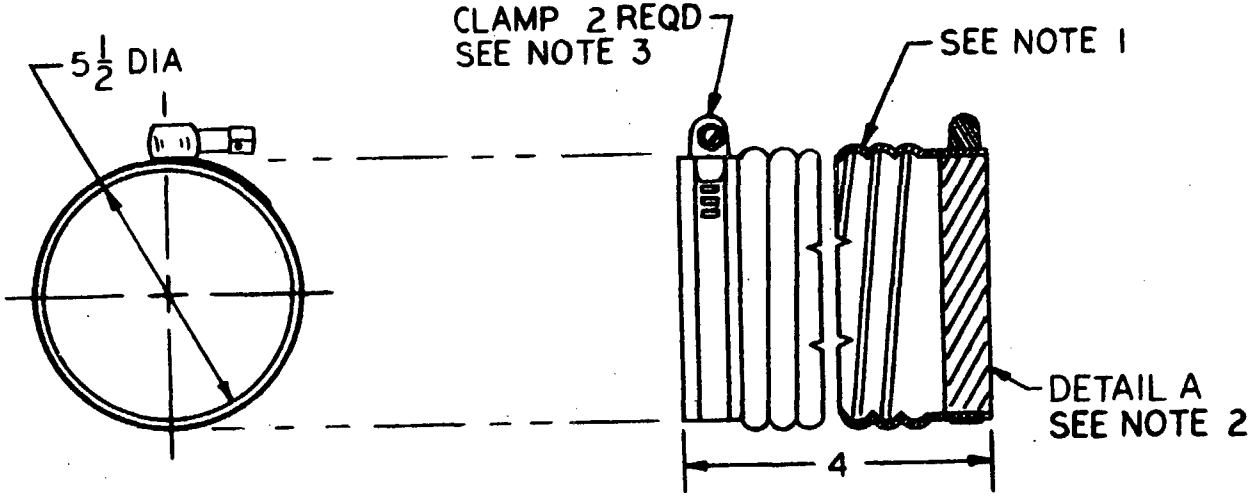
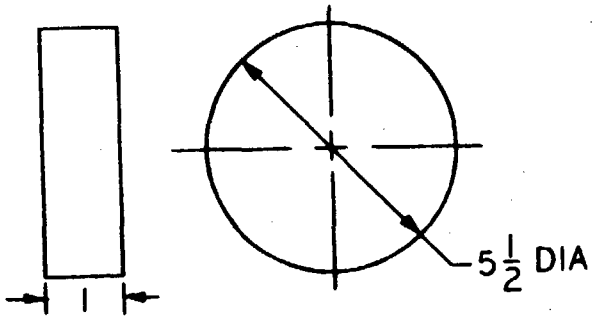



FIGURE 4. Screen.



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DETAIL A-1 REQD

NOTES:

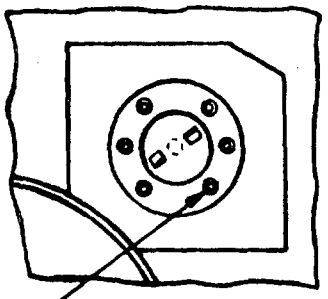
1. HOSE FLEXIBLE TO BE MADE FROM MATERIAL CONFORMING TO MIL-H-52079 OR EQUIVALENT.
2. PLUG TO BE MADE FROM 1 INCH PLYWOOD, NN P 530, STD. INT. WITH EXT. GLUE.
3. USE MS35842-16 OR EQUIVALENT CLAMPS.

FIGURE 5. Boot air restrictor.

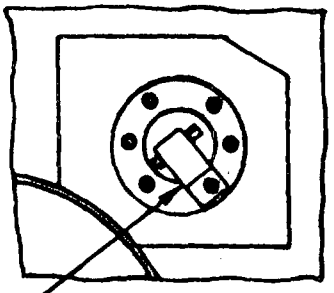




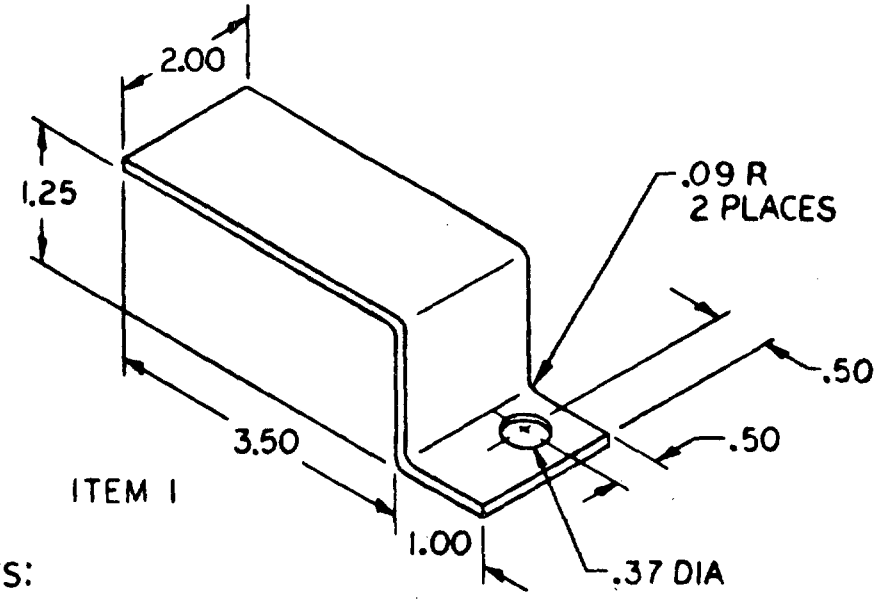
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REMOVE SCREW AND WASHER IN RELATIONSHIP WITH EAR'S OF GAS CAP



PLACE BRACKET ITEM 1 BETWEEN EAR'S OF GAS CAP AND SECURE WITH REMOVED SCREW AND WASHER




NOTES:

1. MATERIAL: STEEL, CARBON, HR. QQ-S-698, .125 THK.
2. FINISH: MIL-C-16173, GRADE 1.

FIGURE 6. Bracket.



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* APPENDIX A

REQUIREMENTS FOR ASSEMBLY, INSTALLATION AND
REMOVAL OF VEHICLE CLOSURE FOR M548 VEHICLE

10. SCOPE

10.1 Scope. This appendix covers requirements for assembly, installation and removal of the vehicle closure (see figure 7).

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this appendix to the extent specified herein.

* SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-P-530 | - Plywood, Flat Panel. |
| QQ-S-741 | - Steel, Carbon, Structural Shape, Plate and Bar. |
| TT-E-527 | - Enamel, Alkyd, Lusterless. |
| TT-P-636 | - Primer Coating, Alkyd, Wood and Ferrous Metal. |

MILITARY

| | |
|---|---|
| MIL-R-3065 | - Rubber, Fabricated Products. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | - Color. |

MILITARY

| | |
|---|---|
| MIL-STD-731 | - Quality of Wood Members for Containers and Pallets. |
| MS16992 | - Bolt (Screw), Lag, Hex Head, Gimlet Point. |



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| | |
|---|---|
| MS27183 | - Washer, Flat-round, Steel, Cadmium Plated, General Purpose. |
| MS51967 | - Nut, Plain, Hexagon-carbon Steel, Cadmium Plated, UNC-2B. |
| MS51968 | - Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNF-2B. |
| MS90725 | - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A. |

* 20.2 Other publications. The following documents form a part of this appendix to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM A576 | - Special Quality Hot Rolled Carbon Steel Bars. |
| ASTM A675 | - Special Quality Hot Rolled Carbon Steel Bars and Bar Size Shapes, Subject to Mechanical Property Requirements. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

30. REQUIREMENTS

30.1 Vehicle closure.

30.1.1 Installation of closure.

a. Install two studs, detail 17, figure 7, into the two 1/2 inch diameter threaded holes in the top of the vehicle side panels at the engine compartment bulkhead.
b. Place closure on vehicle with the 3/4 inch diameter holes at the rear located on the studs installed in step a. The two 5/8 inch diameter holes in the forward support are to be located over the windshield mounting holes.
c. Place a rubber washer, detail 22, figure 7, and a steel washer (MS27183-19), detail 21, figure 7, over each of the studs installed in step a. Install nut (MS51968-14), detail 24, figure 7, on each of the studs and torque securely.



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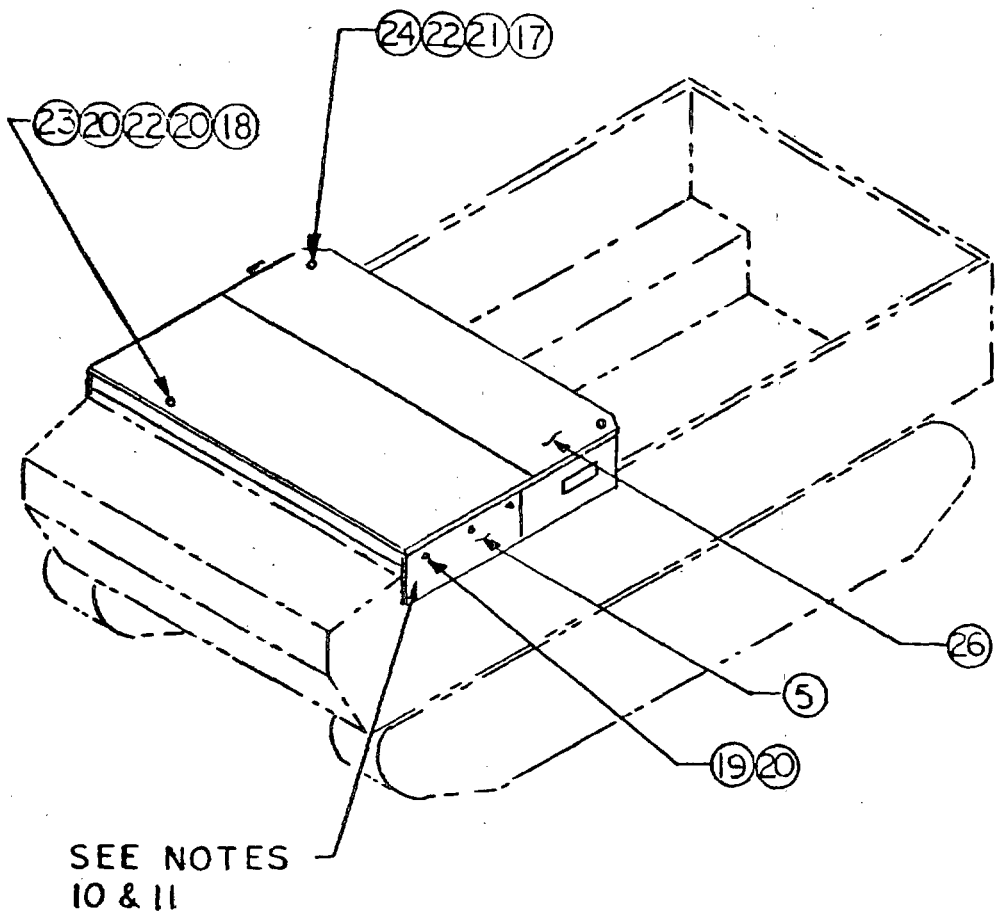
d. Place a steel washer (MS27183-15), detail 20, figure 7, on each of the two screws (MS90725-74), detail 18, figure 7, to be next to the screw head. Place rubber washer, detail 22, figure 7, on each of the screws. Install screws with washers through the 5/8 inch diameter holes in the forward support and through the windshield mounting holes in the vehicle. Install steel washer (MS27183-15), detail 20, figure 7, onto each of the screws. Install nut (MS51967-8), detail 23, figure 7, on each of the screws and torque securely.

30.1.2 Removal of closure.

a. Remove the nuts and washers from the two screws through the closure forward support and the vehicle windshield mounting holes. Withdraw the screws with the remaining washers.
b. Remove the nuts and washers from two studs at the rear of the closure.
c. The closure is now free to be lifted from the vehicle.
d. After removal of the closure, remove the two studs at the rear of the engine compartment.



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Sheet 1 of 6

FIGURE 7. Vehicle closure.



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Sheet 2 of 6

| ITEM | PART NAME | NO. REQ | DESCRIPTION | MATERIAL |
|---|---|---|---|---|
| 1 | TOP FRONT | 1 | 1/2 X 42-1/16 X 101-1/2 | SEE NOTE 1 |
| 2 | TOP REAR | 1 | SEE DETAIL 1 | |
| 3 | BACK PANEL | 1 | 1/2 X 8-1/2 X 93 | |
| 4 | SIDE PANEL | 2 | SEE DETAIL 2 | |
| 5 | SIDE PANEL | 2 | SEE DETAIL 4 | |
| 6 | DEFLECTOR | 2 | SEE DETAIL 3 | |
| 7 | END PANEL | 4 | 1/2 X 2-3/4 X 5 | |
| 8 | BLOCK | 2 | 1/2 X 6-5/8 X 4 | |
| 9 | BLOCK | 2 | 1/2 X 6-5/8 X 7-1/4 | |
| 10 | BLOCK | 2 | 3/8 X 6-5/8 X 4 | SEE NOTE 1 |
| 11 | SUPPORT | 3 | 2 NOM X 2-5/8 X 100-1/2 | SEE NOTE 2 |
| 12 | SPACER | 2 | 2 NOM X 2-5/8 X 23-1/4 | |
| 13 | SPACER | 2 | 2 NOM X 2-5/8 X 15 | |
| 14 | SPACER | 2 | 2 NOM X 2-5/8 X 29-9/16 | |
| 15 | BLOCK | 2 | 2 NOM X 2-5/8 X 2-3/4 | SEE NOTE 2 |
| 16 | SCREEN VENT | 2 | SEE DETAIL 6 | SEE NOTE 3 |
| 17 | STUD | 2 | SEE DETAIL 7 | SEE NOTE 4 |
| 18 | SCREW | 2 | 3/8-16 X 4 UNC-2A | MS90725-74 |
| 19 | BOLT LAG | 6 | 3/8 X 2 STEEL | MS16992-543 |
| 20 | WASHER | 10 | .438 ID X 1.000 O.D | MS27183-15 |
| 21 | WASHER | 2 | .562 ID X 1.375 O.D | MS27183-19 |
| 22 | WASHER | 4 | SEE DETAIL 8 | SEE NOTE 5 |
| 23 | NUT | 2 | 3/8-16 UNC-2B | MS51967-8 |
| 24 | NUT | 2 | 1/2-20 UNF-2B | MS51968-14 |
| 25 | REAR ASSY | 1 | SEE DETAIL 5 | -- |
| 26 | COVER ASSY | 1 | SEE DETAIL 9 | -- |

FIGURE 7. Vehicle closure.- Continued



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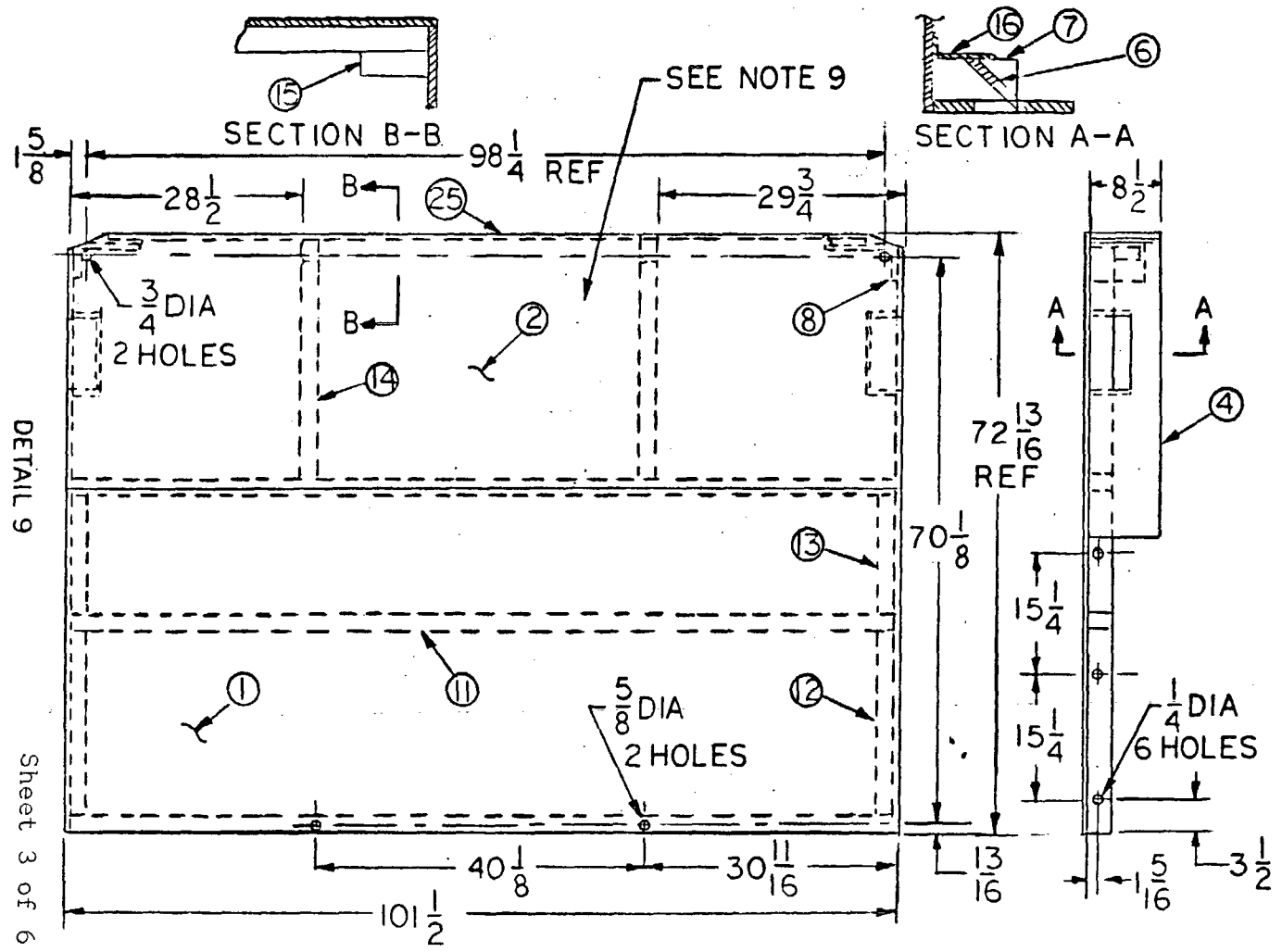
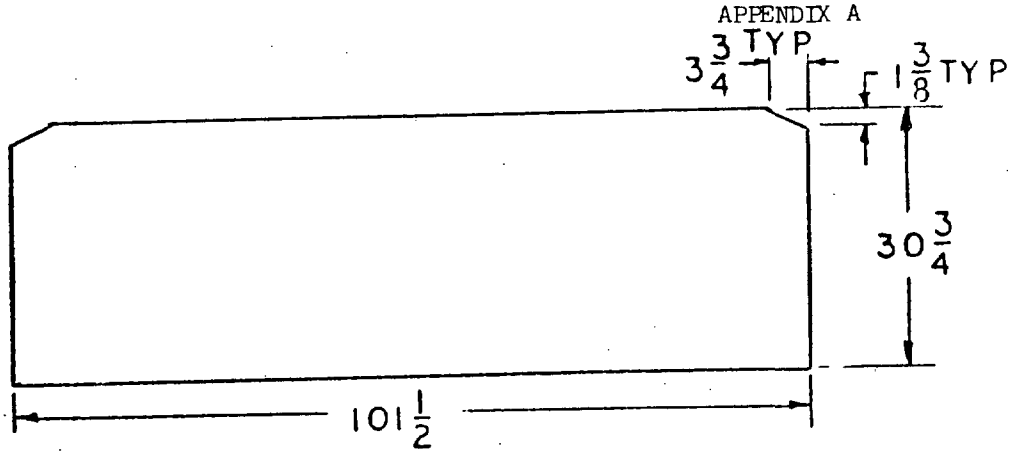



FIGURE 7. Vehicle closure.-Continued

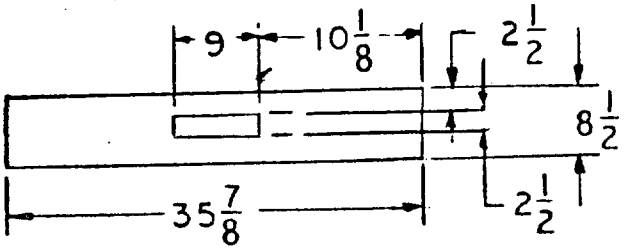
Sheet 3 of 6



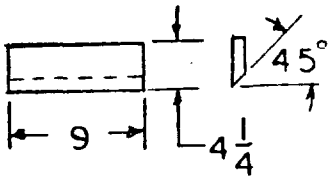
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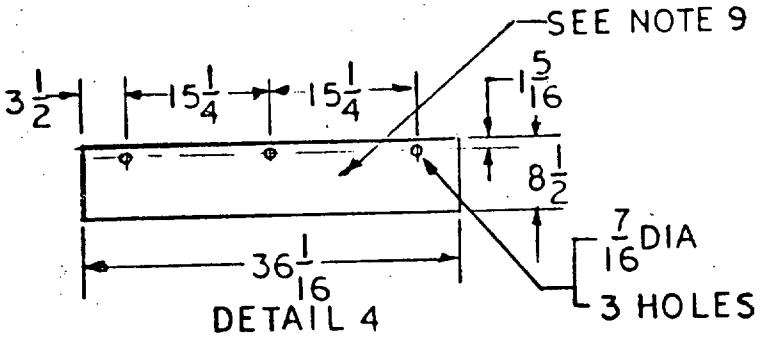

DETAIL 1



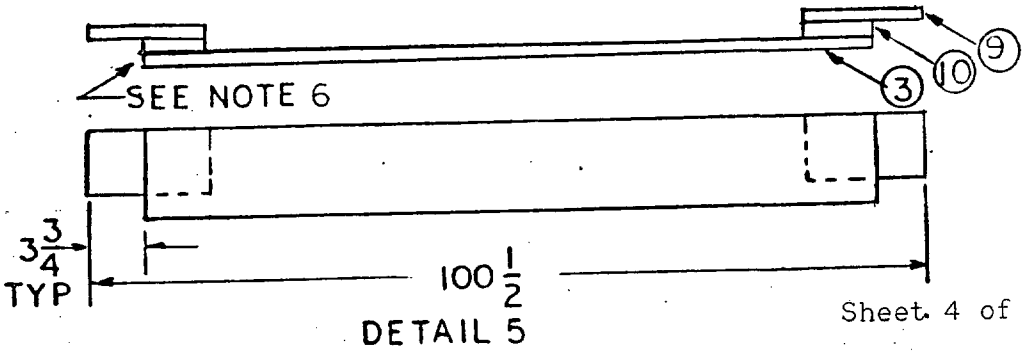

DETAIL 2




DETAIL 3




DETAIL 4



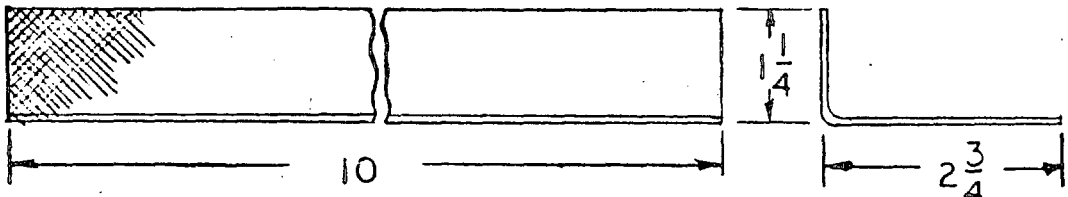

DETAIL 5



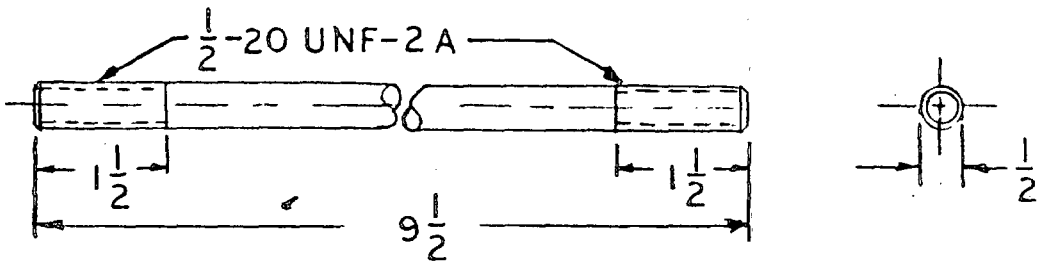
FIGURE 7. Vehicle closure.- Continued



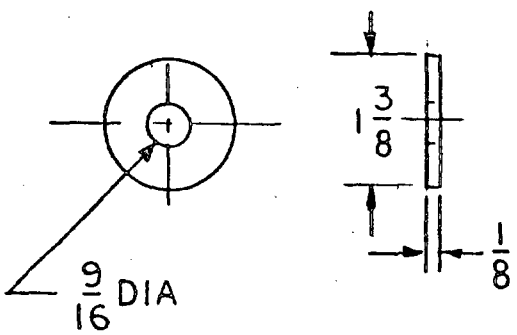
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DETAIL 6
MATERIAL: NOTE 3




DETAIL 7
MATERIAL: NOTE 4




DETAIL 8
MATERIAL: NOTE 5



FIGURE 7. Vehicle closure.-Continued



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NOTES:

1. PLYWOOD, SPEC NN-P-530, EXTERIOR TYPE, PS 1-66 B-C EXT-DFPA(4)

2. LUMBER, SPEC MIL-STD-731, GROUP I OR II

3. WIRE CLOTH, STEEL CARBON 0.0348 DIA 4 X 4 MESH TYPE I, CLASS I, SPEC RR-W-360

4. STEEL, CARBON, SPEC QQ-S-741, OR ASTM A576, STEEL CARBON, OR ASTM, A675

5. RUBBER, GRADE SB515 $A_1$ $B_1$ $C_1$, SPEC MIL-R-3065

6. ITEMS 3, 9 & 10 SHALL BE LAMINATED TOGETHER WITH WATERPROOF GLUE

7. ALL UNPAINTED METAL PARTS (EXCEPT NAILS) SHALL BE COATED WITH MIL-C-16173, GRADE 1

8. NAILING SHALL CONFORM TO SPEC PPP-B-621

9. FINISH EXTERIOR SURFACES

   CLEAN PER PROCESS C-1, MIL-P-116

   APPLY PRIMER, CMPSN OPTL, SPEC TT-P-636 0.75 TO 1.25 MILS THICK

   APPLY ENAMEL, OLIVE DRAB, NO. 34087, CMPSN OPTL, SPEC TT-E-527

10. PAINT USED FOR MARKING SHALL CONFORM TO SPEC TT-E-527, COLOR YELLOW NO. 33538

11. STENCIL ON REMOVABLE PANEL AT LEFT DOOR "TO SET OR RELEASE BRAKES - REMOVE THIS PANEL"



FIGURE 7. Vehicle closure.-Continued.



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* APPENDIX B

REQUIREMENTS FOR FABRICATION, ASSEMBLING
LEVEL B VEHICLE CLOSURE FOR M548 VEHICLE

10. SCOPE

10.1 Scope. This appendix covers requirements for assembly and installation of the vehicle closure (see figure 8).

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this appendix to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-P-530 | - Plywood, Flat Panel. |
| PPP-S-760 | - Strapping, Non-metallic (and Connectors). |

MILITARY

| | |
|---|---|
| MIL-B-13239 | - Barrier Material, Waterproofed, Flexible, All Temperatures. |

* STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-731 | - Quality of Wood Members for Containers and Pallets. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods. |

30. REQUIREMENTS

30.1 Heater intake grille. Heater grille shall be covered with MIL-B-13239, Type CW-2. The barrier material shall cover the entire grille and shall extend three inches over each side. Secure barrier material in place with tape in accordance with 30.4 (see figure 8).



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30.2 Engine exhaust grille. Engine exhaust flange shall be sealed with Type IV, Class 1, PPP-T-60 tape or Type II, MIL-T-22085. Grille shall be covered with MIL-B-13239, Type CW-2. The barrier material shall cover the entire grille and shall extend three inches over each side. The barrier material shall have a slit applied at the exhaust flange area. Secure barrier material in place with tape in accordance with 30.4 (see figure 8).

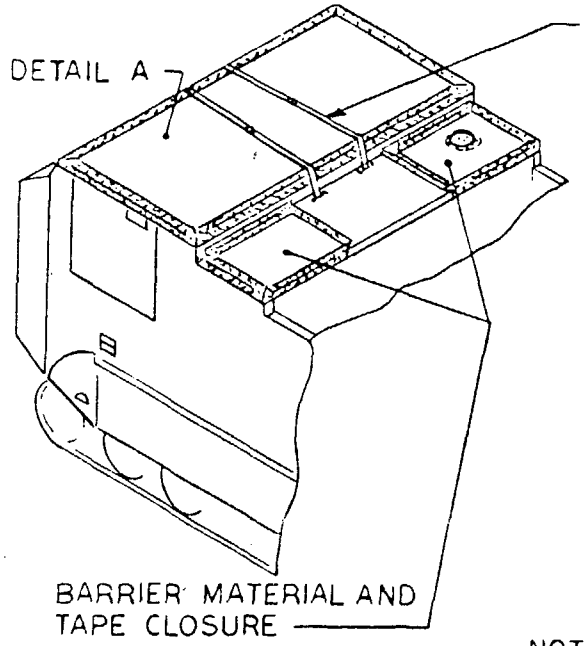
30.3 Driver's compartment closure. The driver's area shall be covered with a single piece of plywood conforming to figure 8, secured in place with tape in accordance with 30.4. Plywood closure shall be secured down with two non-metallic straps conforming to PPP-S-760 as shown in figure 8.

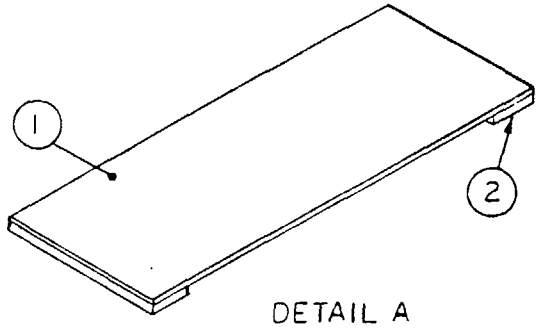
30.4 Taping requirements. Tape shall conform to Type IV, Class 1, PPP-T-60, or Type II, MIL-T-22085. Tape shall be 3 inches in width and applied in such a manner that a minimum of wrinkles occur. When, because of turns or irregular surfaces, wrinkles occur, the tape shall be cut through the wrinkles, and the cut edges overlapped. Tape shall be applied in longest possible lengths to minimize overlapping. Extreme care shall be exercised in applying tape to eliminate wrinkles and loose edges. See figure 8 for taped areas.



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INDICATES TAPED AREAS




DETAIL A

| ITEM | DESCRIPTION | REQD. | NOTE |
|---|---|---|---|
| 1. | 1/2 X 42-1/16 X 100 | 1 | 1 |
| 2. | NOM 1 X 4 X 42-1/16 | 2 | 2 |

NOTES:

1. PLYWOOD, SPEC NN-P-530, STD INT W/EXT GLUE.
2. LUMBER, SPEC MIL-STD-731, GROUP I OP II.
3. NAILING SHALL CONFORM TO MIL-STD-1186.

FIGURE 8. Vehicle closure.



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